



Technical Campus

योग: कर्मसु कौशलम्
IN PURSUIT OF PERFECTION



**SCHOOL OF ENGINEERING
AND TECHNOLOGY**

Process Automation (IOT310)

By: Dr. Divya Agarwal

Course Overview:

- Process and Industrial Automation specialization offers comprehensive knowledge and professional level skills focused on developing and deploying software robots. It starts with the basic concepts of Robotic Process Automation. In the present global scenario, industries are moving towards complete automation. Hence, this course is foundation of engineers who want to specialize in industrial and process automation field.

Course Objective:

- To understand the Fundamental and basic principles of Robotics Process Automation, Applications in various Industries.
- To install the UiPath Studio and understand difference between different versions
- To use different activities while using excel sheet and automate email related activities.
- To understand benefits of given industrial automation systems, describe their functions and compare characteristics of given automation systems.

Course Outcomes (CO):

- **CO1** Able to Map and assess some of the business processes that are fit for automation.
- **CO2** Able to connect to UiPath Automation Cloud, download the installer, and set up their own attended automation environment and Build an automation using StudioX.
- **CO3** Able to effectively automate tasks involving use of Excel files, adapt the automation project to dynamic Excel ranges.
- **CO4** Able to apply and map realtime application of industrial automation

Syllabus Contents

- **UNIT III**
- **Error Handling:** Use StudioX tools to validate and analyze automation projects for handling and troubleshooting errors, build automations using best practices that increase reusability and readability.
- **Email Automation:** Use actions and resources related to email automation, create automation projects using StudioX email specific activities, create email content using both text and HTML options.
- **Excel Automation:** Automate tasks involving use of Excel files such as usage of cell activities, range activities, pivot activities, chart activities and workbook activities.

Syllabus Contents



- **UNIT IV**
- **Introduction to Industrial Automation:** Need and benefits of Industrial Automation, Automation Hierarchy, Basic components of Automation systems, description of each component, types of automation systems flexible, fixed and programmable systems
- **The Future Automated Factory:** Future Automated Factory, Human Workers in the Future Automated Factory, The social impact

Syllabus Contents

- **UNIT III**
- **Error Handling:** Use StudioX tools to validate and analyze automation projects for handling and troubleshooting errors, build automations using best practices that increase reusability and readability.
- **Email Automation:** Use actions and resources related to email automation, create automation projects using StudioX email specific activities, create email content using both text and HTML options.
- **Excel Automation:** Automate tasks involving use of Excel files such as usage of cell activities, range activities, pivot activities, chart activities and workbook activities.

What is error handling?

- Error handling is a process in programming that deals with unexpected situations (errors) that arise during the execution of a program.
- **Why it occurs:** Errors can happen due to various reasons:
 - **User mistakes:** Entering incorrect data, typos in commands.
 - **System issues:** Hardware malfunctions, software bugs.
 - **External factors:** Network problems, unexpected data formats.

Importance of Error Handling

- **Prevents data loss and wasted efforts:** By gracefully handling errors, StudioX automation ensures the process doesn't crash unexpectedly, potentially leading to lost data or incomplete tasks.
- **Enables informed decision making:** Error messages provide valuable information about the issue, allowing the user to understand the cause and take appropriate action.
- **Improves automation maintainability:** A well-designed automation with error handling logic is easier to troubleshoot and adapt to future changes.

StudioX Approach to Error Handling

- **User intervention:** StudioX relies on user interaction to address errors.
- **Error messages and options:** When an error occurs, StudioX displays a message explaining the issue and offers options like:
 - **Retry:** Allows the user to fix the error and try again.
 - **Skip:** Skips the problematic step and continues the automation.
 - **Stop:** Halts the entire automation process.

1. Identifying Potential Errors:

- **Analyze process:**

- Carefully examine automation workflow to identify potential points of failure.
- Such as element not found, crashed applications, or unexpected data encountered.

- **Review activity properties:**

- Look for built-in error handling mechanisms.
- Look for properties like "Ignore RPA Exceptions" or "Retry" within activity settings.

2. Validating and Analyzing with StudioX Tools:

- **Validation Panel:** Utilize the built-in validation tool to identify potential issues before running the automation. This can reveal missing dependencies, incorrect selectors, or logical errors in the workflow.
- **Code Review:** Manually inspect the workflow visually. Look for proper element identification, logical flow, and error handling mechanisms.

3. Implementing Error Handling Practices:

- **Try-Catch Blocks:** Encapsulate critical sections of your automation within a "Try-Catch" block. The "Try" section contains the main automation logic, while the "Catch" section handles any exceptions that might occur.
- **Re-Try Mechanism:** For certain errors (e.g., network issues), you can implement a retry mechanism within the "Catch" block. This allows the automation to attempt the action again a specified number of times before raising a final error.

4. Debugging Techniques:

- **Breakpoints:** Set breakpoints at specific points in the workflow to pause execution and examine variable values or element states. This helps pinpoint the exact location where errors occur.
- **Logging:** Utilize the "Log Message" activity to record information about the automation's execution. This can be helpful for tracing the flow and identifying issues during troubleshooting.

5. Building Reusable and Readable Automations:

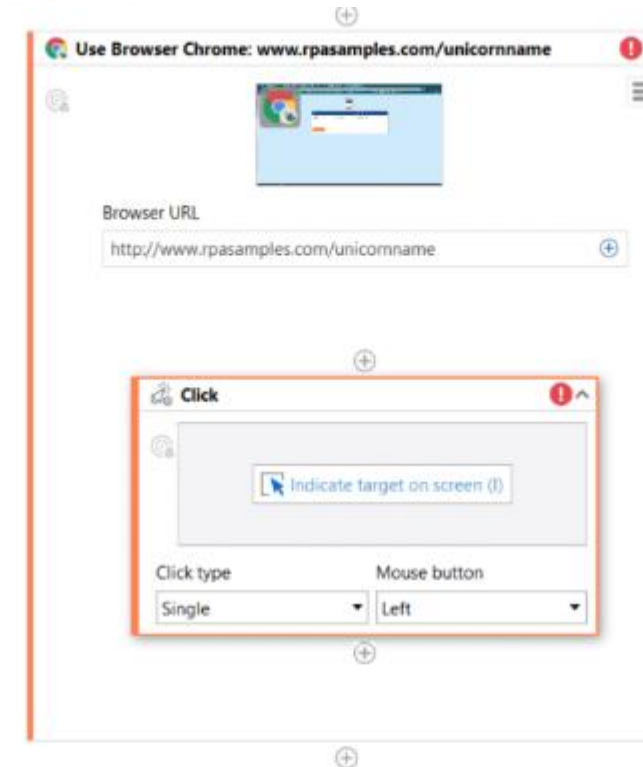
- **Modular Design:** Break down complex workflows into smaller, reusable functions or sub-processes. This improves code organization and simplifies maintenance.
- **Meaningful Variable Names:** Use descriptive names for variables to enhance code readability and understanding.
- **Error Handling Comments:** Document the implemented error handling mechanisms using comments within the workflow. This clarifies the purpose of each step and aids future modifications.

Error Handling Tools

- StudioX comes with two tools that will help you anticipate some errors.
- **Validate** and **Analyze**
- Both found under the **Analyze** dropdown.
- Validate and Analyze options check your workflow for incomplete activities or against the organization's rules.
- These options do not replace your capabilities to build coherent, reliable projects.

Validate

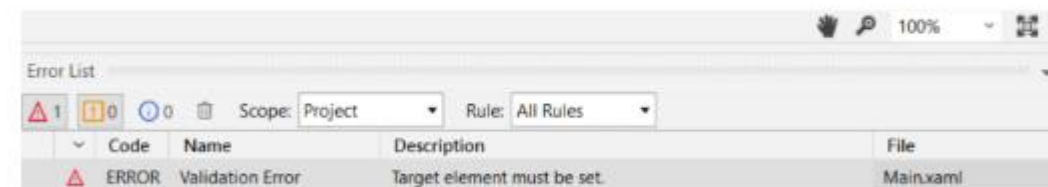
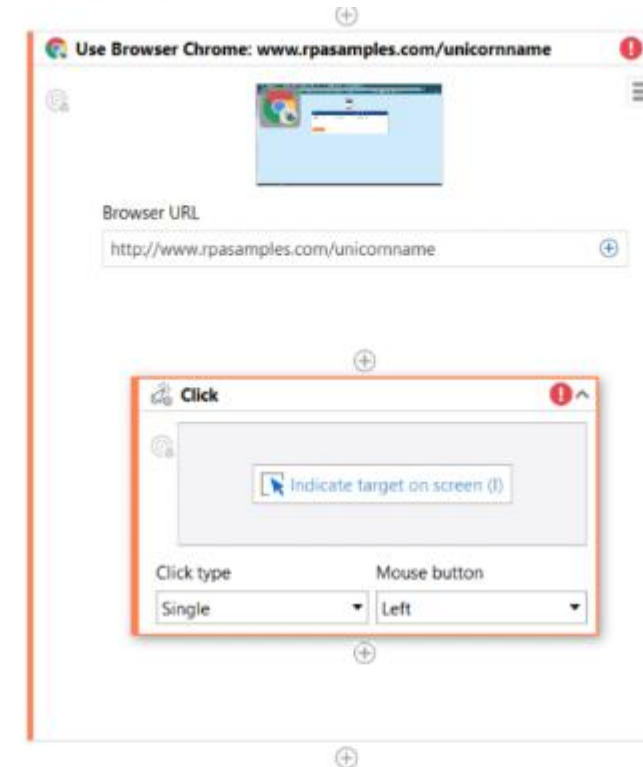
- Validate tool will find all the activities with validation errors.
- These are usually incomplete activities.
- Can easily spot them as having the red icon in the upper-right corner.
- Cannot run or publish a project until all the validation errors are solved.

A screenshot of the 'Error List' panel in the software development tool. It shows a table with error details. The table has columns for 'Code', 'Name', 'Description', and 'File'. There is one error listed: 'ERROR Validation Error Target element must be set.' in the file 'Main.xami'. The panel also includes a search bar, a 'Scope' dropdown set to 'Project', and a 'Rule' dropdown set to 'All Rules'. There are also icons for error types (red triangle, yellow square, blue circle) and a count of errors (1, 0, 0).

Code	Name	Description	File
ERROR	Validation Error	Target element must be set.	Main.xami

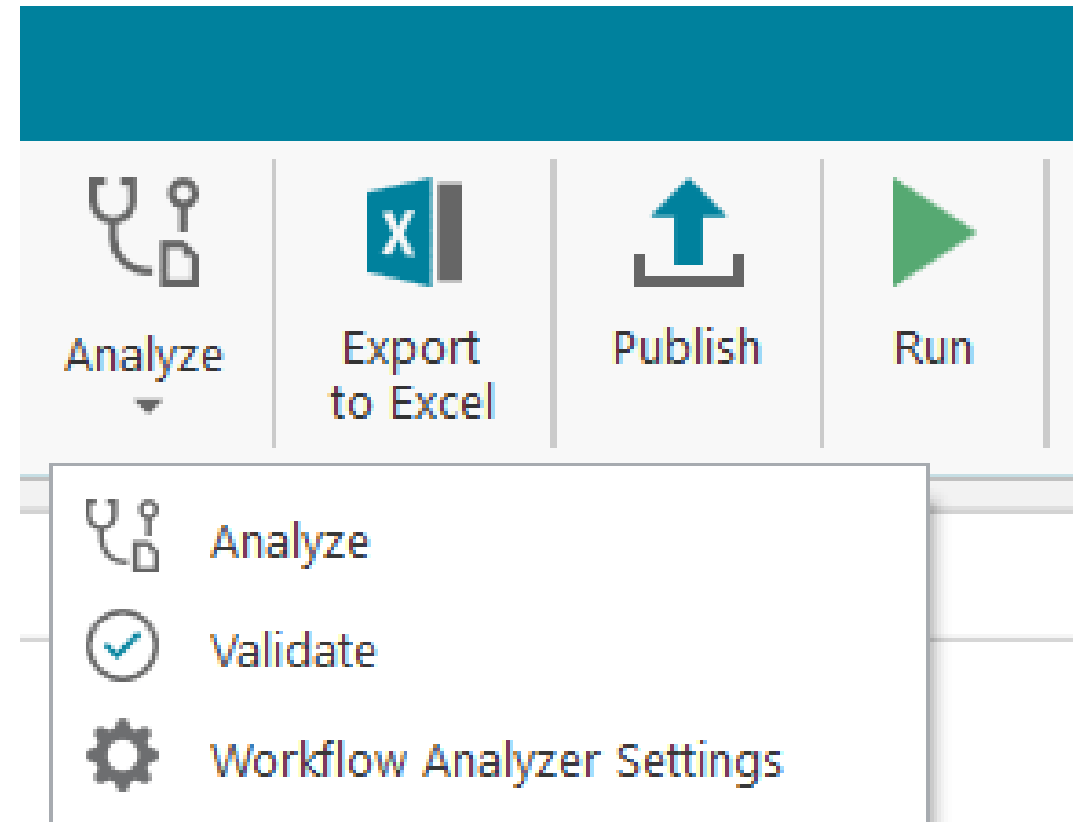
Validate

- Notice in screenshot, both resource and action have a red icon, even though resource has all the information provided.
- Open error list panel which has information about the error.
- It says that "target element must be set".
- Double-clicking on it will take you to the activity with validation errors, and in this example, you can see that the target for the Click action was not specified.



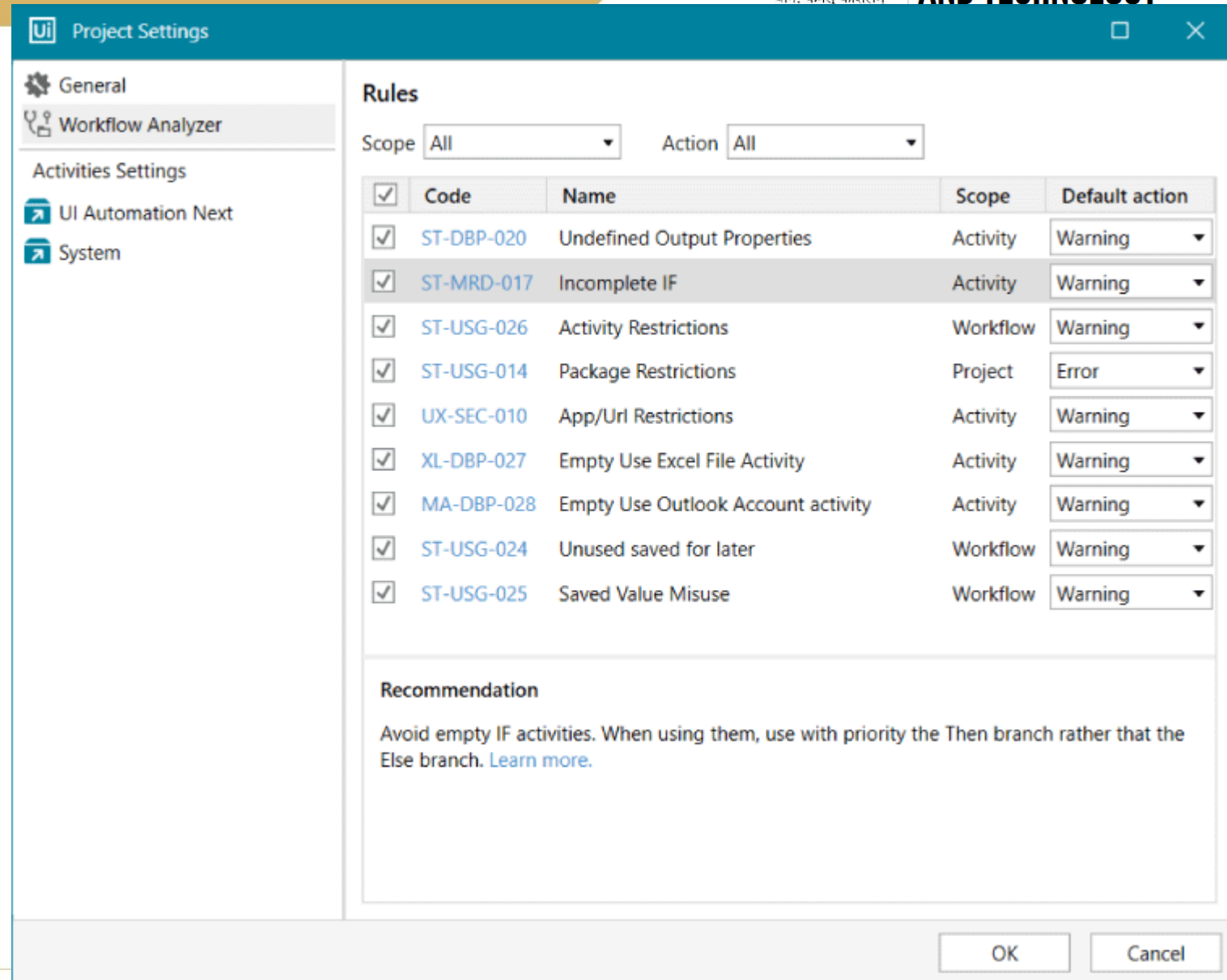
Analyze

- StudioX also comes with a set of built-in best practices and analyzes the project against them.
- Access Workflow Analyzer options by clicking **Analyze** drop-down menu from Design ribbon tab.
- **Analyze** option, checks the workflow against a set of predefined rules;
- **Workflow Analyzer Settings** option, allows you to manage, enable, or disable rules.



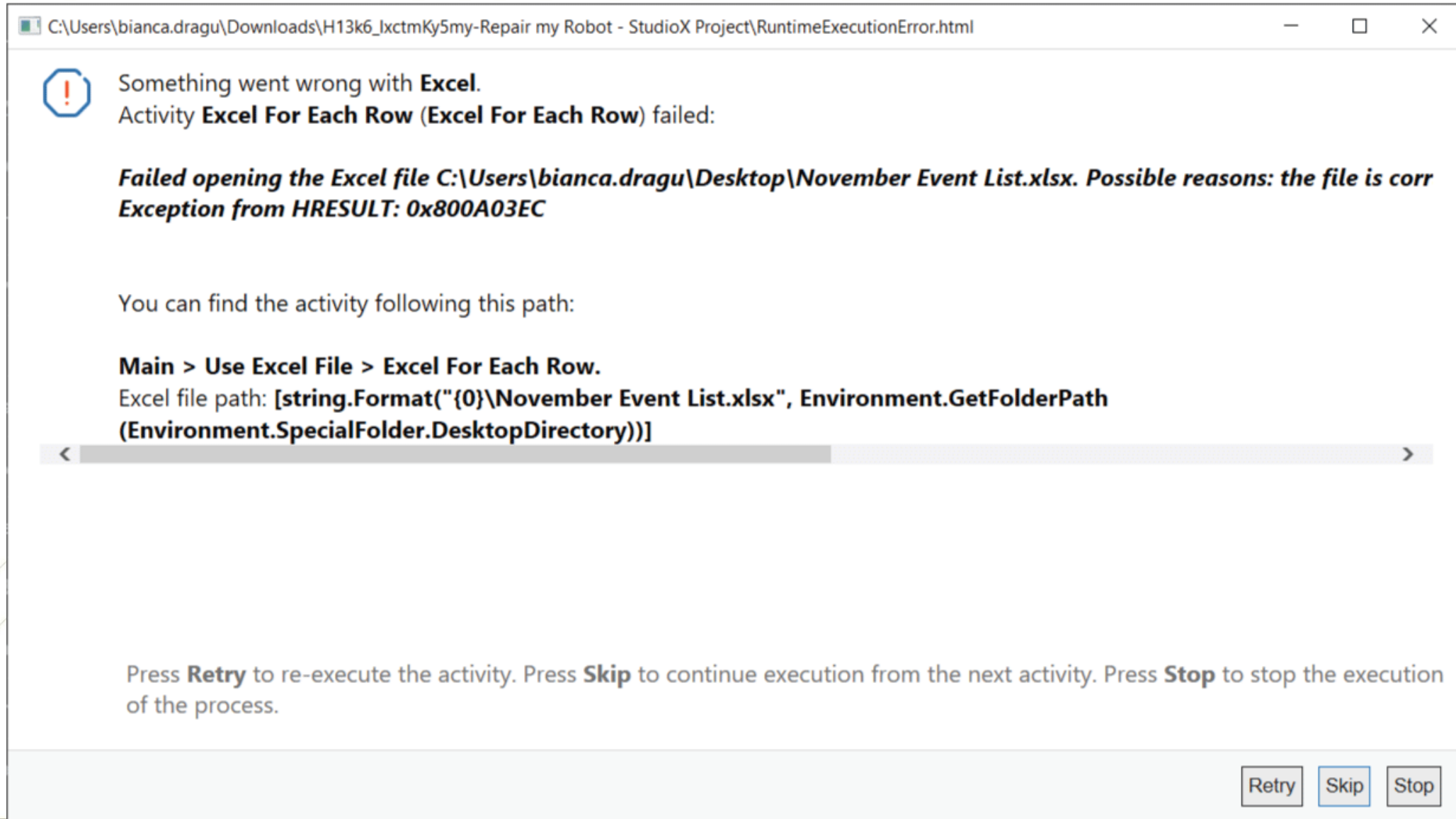
Built-in Rules

- StudioX comes with a set of built-in rules, enabled by default, that check the activities, the workflow, and the whole project. To examine them, select the **Workflow Analyzer Settings** option.



- **"Activity Restrictions"**, **"Package Restrictions"** and **"App/Url Restrictions"** are managed by an organization and they enforce limitations on the activities, websites, apps, and packages used in automation.
- **"Empty Use Excel File"** and **"Empty Use Outlook Account"** check if you provided an action to use with the resource.
- **"Incomplete IF"** checks if at least one branch (the Then branch is recommended) contains activities.
- If you **Save For Later Use** and don't use the value, the Analyzer will throw a warning.
- **"Undefined Output Properties"**. This rule checks whether output properties for certain activities are declared and used.

Error message



- Create **Robot Path**
- Break it down into a sequence of steps to identify easier actions and resources needed to use.
- **Rename Activities** in order to have more specific names for them or do not use the same name for multiple activities (Helps to identify problematic activity in an Error Message)

Troubleshooting Practices

- For eg: One 'Read CSV' action can be **renamed** in order to specify file that should be read.
- This way, when using output for later use it will be easier to identify the content.

Read CSV

Read from file

Report_February2020.csv

Output to

[x] [Report_February] Report_February

☒ Has headers

Read CSV

Read from file

Report_March2020.csv

Output to

[x] [Report_March] Report_March

☒ Has headers

Read CSV: Report February

Read from file

Report_February2020.csv

Output to

[x] [Report_February] Report_February

☒ Has headers

Read CSV: Report March

Read from file

Report_March2020.csv

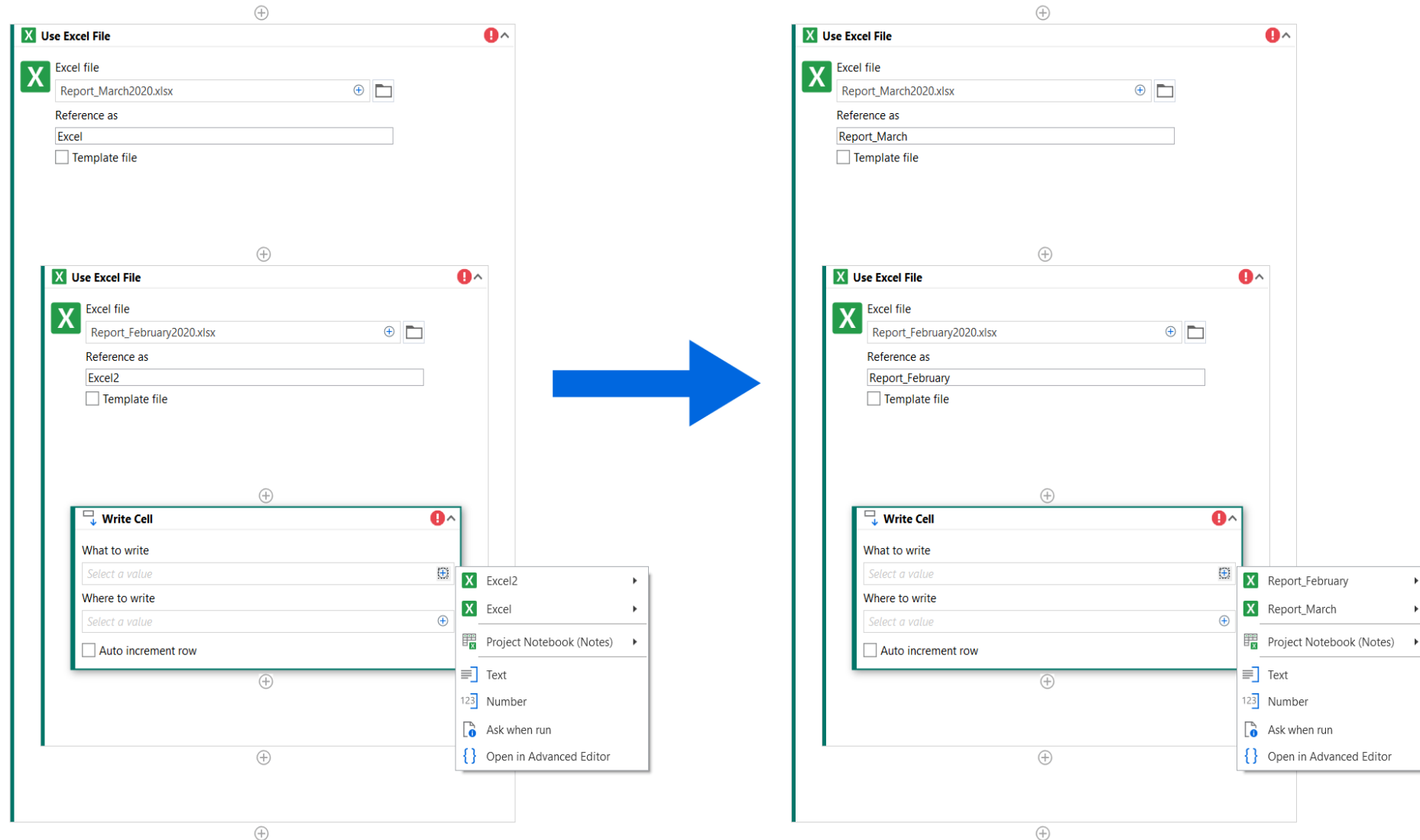
Output to

[x] [Report_March] Report_March

☒ Has headers

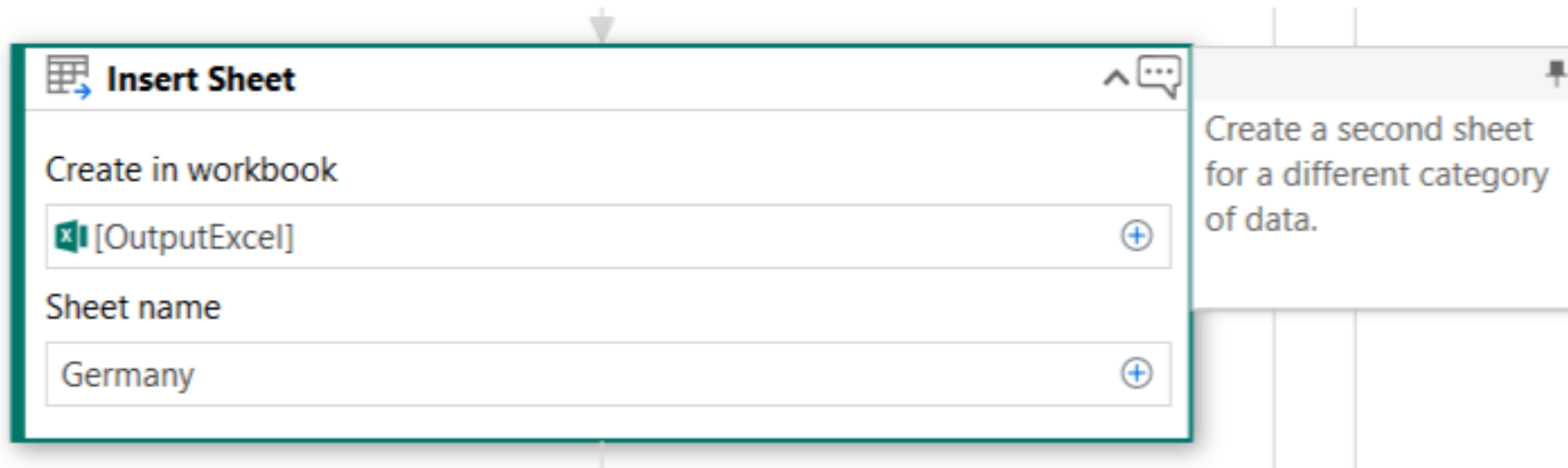
Troubleshooting Practices

- For all the Activities that have the option of 'Reference as' (such as Use Excel File resource, Use Outlook Account resource, etc.) use **suggestive naming**.



Troubleshooting Practices

- Use **Annotations** for Activities in order to add more information for documentation purposes. This is especially helpful when sharing a project with another person.

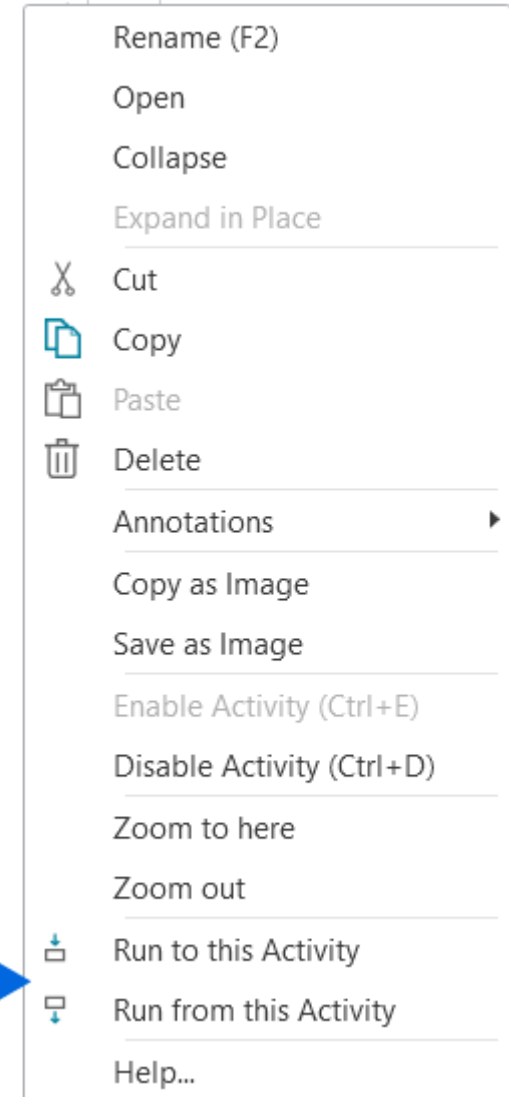


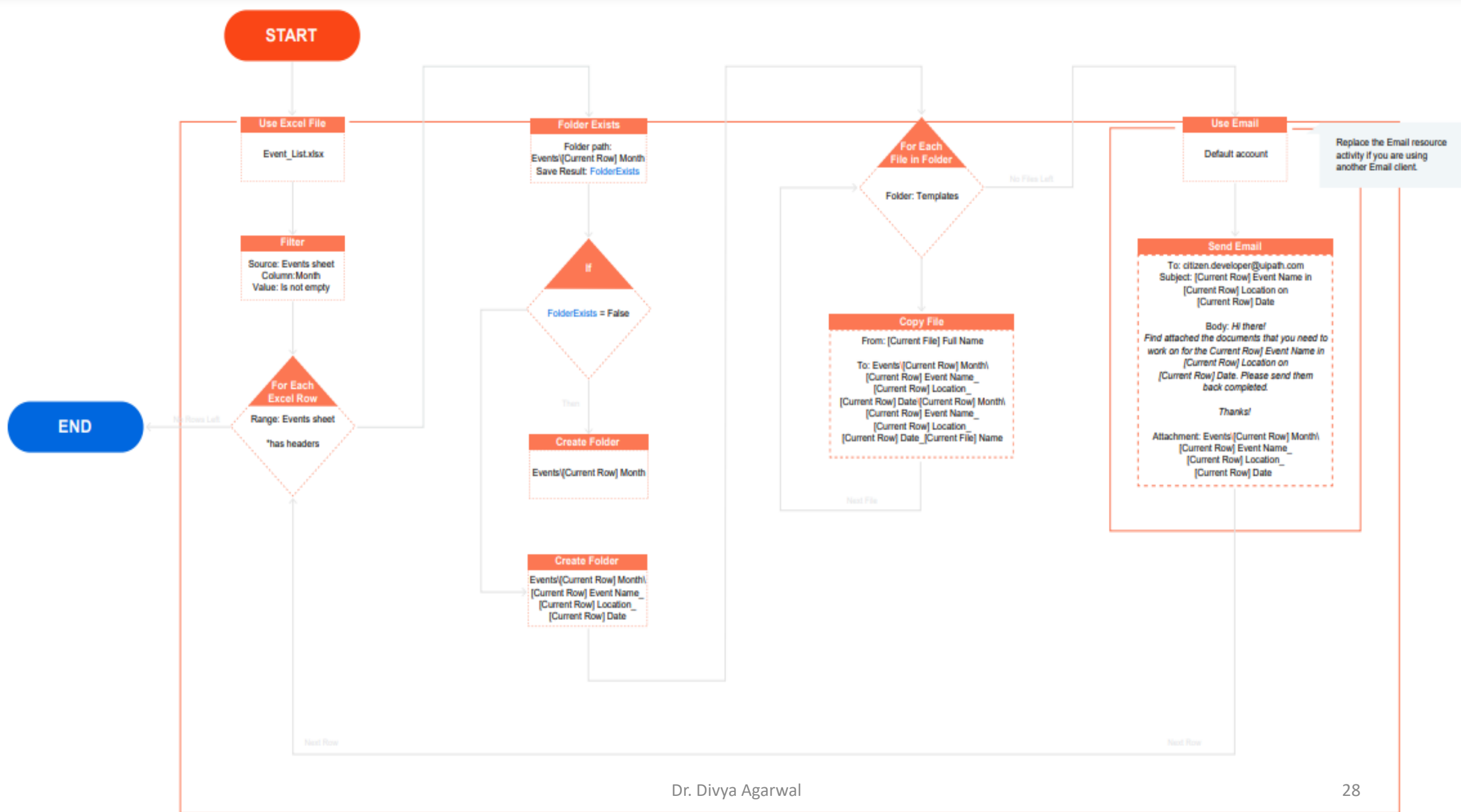
Troubleshooting Practices

- **Close all Excel applications** before running a project.
- Use the local **Project Notebook** for any Data Manipulation, but do not save any values in it.
- Constantly check for the **input data** to be correct. At the beginning of the process, and also in the development phase.
- Use “**Message Box**” **Activities** when creating a Task Automation before and after each important step/block of your automation. (i.e. write a certain message such as "Started reading file").

Troubleshooting Practices

- As you build your project, test small parts of it often.
- There are actions that do not overwrite, for example, **Create Pivot Table**, which means that a second run might result in an error.
- Make use of the **Run to this Activity**, to run the project from the start and stop before a specific activity, or **Run from this Activity** to run the project starting with a specific activity.
- You can find these options in the Context Menu (right-click on the activity).





ERROR HANDLING TUTORIAL

- <https://youtu.be/sUjWKiQ0JJU?t=469>

EMAIL AUTOMATION

- Email clients are most heavily used business applications.
- Capable of doing far more than simply sending and receiving electronic communications, email clients are tightly integrated into a variety of office productivity tools, giving tools like Gmail and Outlook the ability to update contacts, schedule calendar appointments, share files with team members, and even assign tasks to coworkers.
- As a business user, a great deal of time is spent on sending and receiving email messages, repeatedly working with specific email groups, and downloading important attachments on a daily basis.

What is Email Automation?

- Email automation refers to task automation related to your email account.
- Email automation involves using software to automatically send personalized emails based on pre-defined triggers or schedules.
- This eliminates the need for manual mass emailing and streamlines communication.



What is Email Automation?

- For Example:
 - Automate minor tasks involving your email account that have a predictable structure, or those that are part of larger, more complex automation projects.
 - Sending personalized welcome emails to new subscribers.
 - Triggering automated responses based on user actions (e.g., order confirmations).
 - Sending follow-up emails based on specific criteria.
 - **Birthday or Anniversary Emails:** Personalized messages to celebrate special occasions with customers

Why is Email Automation Important

- **Saves Time:** Automates repetitive tasks, freeing up your time for more strategic work.
- **Increased Efficiency:** Sends targeted emails at the right time, improving campaign performance.
- **Improved Personalization:** Delivers relevant content to specific audiences, boosting engagement.
- **Scalability:** Manages large email lists and campaigns effortlessly.

How to use Email Automation?

- When creating a new project that involves the automation of Gmail or Outlook, you will use one of the following three email automation resources:
- The **Use Desktop Outlook App** activity
- The **Use Outlook 365** activity
- The **Use Gmail** activity

Email Automation?

UiPath Mail Activities Pack: UiPath.Mail.Activities

This activity package is designed to facilitate the automation of any mail-related tasks, covering various protocols, such as IMAP, POP3 or SMTP. UiPath also features activities that are specialized for working with Outlook and Exchange.

To better understand the usability of the mail activities in the package, let's take a high-level look at how we can automate the distribution of email alerts generated by multiple servers.



UiPath.Mail.Activities by UiPath

Retrieve and send mail using POP3, IMAP, SMTP and Exchange protocols

Activities available Email Automation

Save Mail Message

Save Attachments

Send SMTP Mail Message

Get POP3 Mail Message

Get IMAP Mail Messages

Move IMAP Mail Message

Get Outlook Mail Messages

Move Outlook Mail Message

Send Outlook Mail Message

Save Outlook Mail Message

Delete Outlook Mail Message

Reply To Outlook Mail Message

Set Outlook Mail Categories

Mark Outlook Mail As Read/Unread

Outlook Mail Messages Triqer

Exchange Scope

Delete Exchange Mail Message

Get Exchange Mail Messages

Move Exchange Mail Message

Send Exchange Mail Message

Delete IBM Notes Mail Message

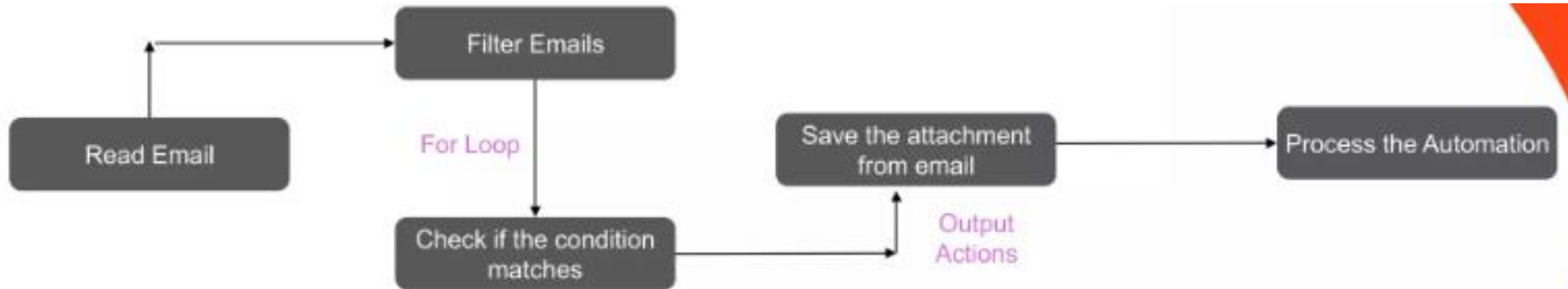
Get IBM Notes Mail Messages

Move IBM Notes Mail Message

Send IBM Notes Mail Message

Create HTML Content

Example: Email Automation



UiPath StudioX - NewBlankTask5

HOME DESIGN

Save Cut Copy Paste Project Project Notebook Manage Packages App/Web Recorder Data Extraction Analyze Export to Excel Publish Run

Activities Search activities (Ctrl+Alt+F)

Resources

- Use Outlook Account Select the Outlook account to...

Actions

- Archive Outlook Email Archives an Outlook email.
- Delete Outlook Email Deletes an Outlook email.
- For Each Email Repeats the contained activitie...
- Forward Outlook Email Forward email to new recipien...
- Mark Email As Read Marks the specified mail mess...
- Move Outlook Email Moves the specified email to a...
- Reply To Outlook Email Reply to an email.
- Save Email Attachments Saves the email message attac...

Main

Expand All Collapse All

Use Outlook Account

Account

Default Email Account

Reference as

Outlook

Drop Activity Here

Properties Data Manager

Output Error List

Orchestrator Not Connected

Step1: Create a Temp folder on your computer and a Search Folder in Outlook.

- Create a folder named **Temp** at the root of the C: drive on your computer.
- Open Outlook, then right-click **Search Folders** in the Folder pane and select **New Search Folder**.
- In the New Search folder window, select **Create a custom Search Folder > Choose**.
- In the Custom Search Folder window, enter "Last week's attachments" in the **Name** box and then select **Criteria**.
- In the **Messages** tab of the Search Folder Criteria window, from the drop-down menus next to **Time** select **received** and **Last Week**.
- From the **More Choices** tab, select **Only items with** and then **one or more attachments** from the adjacent drop-down menu.
- Click **OK** three times to close the windows and save the search folder.

Step 2: Set up the project: Create a new blank project using the default settings.

Step 3: Add the Outlook account to the project.

- In the Activities panel, select the Mail tab and then double-click Use Desktop Outlook App. The activity is added to the Designer panel.
- In the activity, the default email account is already selected in the Account field. If you want to use a different account, select it from the drop-down menu.
- In the Reference as field, leave the default value Outlook as the name by which to refer to the account in the automation.

Step 4: Iterate through the messages in the Outlook folder.

- In the Activities panel, drag the **For Each Email** activity and drop it inside the Use Desktop Outlook App activity.
- In the **For Each Email** activity, click **Plus** on the right side of the **In emails from** field, and then select **Outlook > [Search Folders] > [Last week's attachments]** to select the search folder created in the account Outlook. Your selection is displayed in the **In emails from** field as Last week's attachments.

Iterating Through Outlook Mails and Saving Attachments

1 Use Outlook Account

Account

Default Email Account

Reference as

Outlook

1.1 For Each Mail Message

For each

CurrentMail

In emails from

{ } Last week's attachments

Limit emails to first

100

Additional Filters

☐ Unread mail

☒ Retrieve attachments

☐ With attachments only

☐ Include subfolders

Drop Activity Here

Step 5: Save the attachments for each message to a folder on your computer.

- In the Activities panel, drag the [Save Email Attachments](#) activity and drop it inside the **For Each Email** activity.
- In the Save Email Attachments activity: Click **Plus** on the right side of the **Email** field, and then select **CurrentMail** to indicate that the email whose attachments you want to save is the current email you are iterating through.
- Click **Browse** next to the **Save to folder** field, and then browse to and select the folder on your computer where to save the attachments. For this example, we created a folder called Temp on the C drive of the computer.

Iterating Through Outlook Mails and Saving Attachments

1 Use Outlook Account

Account

Default Email Account

Reference as

Outlook

1.1 For Each Mail Message

For each

CurrentMail

In emails from

Last week's attachments

Limit emails to first

100

Additional Filters

☐ Unread mail

☒ Retrieve attachments

☐ With attachments only

☐ Include subfolders

1.1.1 Save Email Attachments

Email

CurrentMail

Filter by file name (e.g. *.xls*)

Select a value

☒ Exclude Inline Attachments

Save to folder

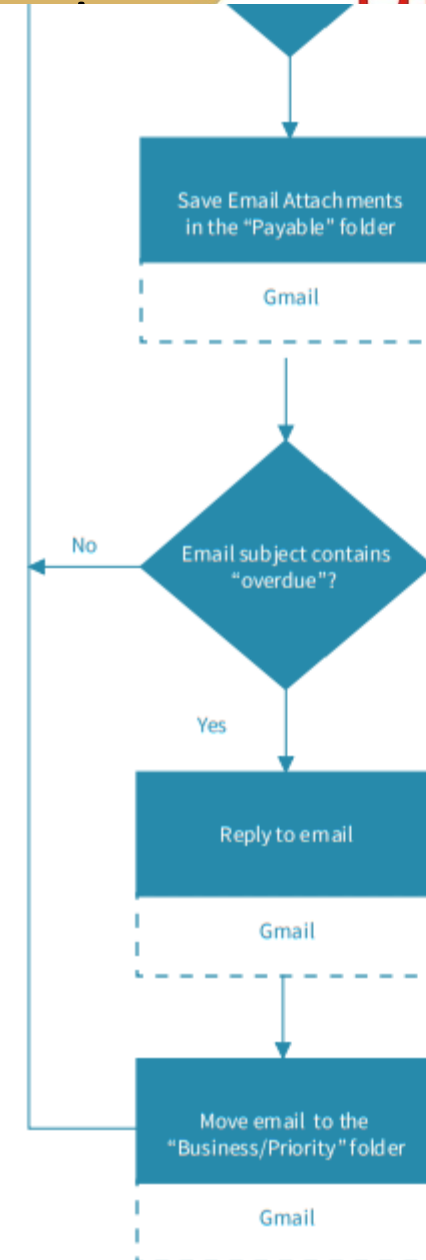
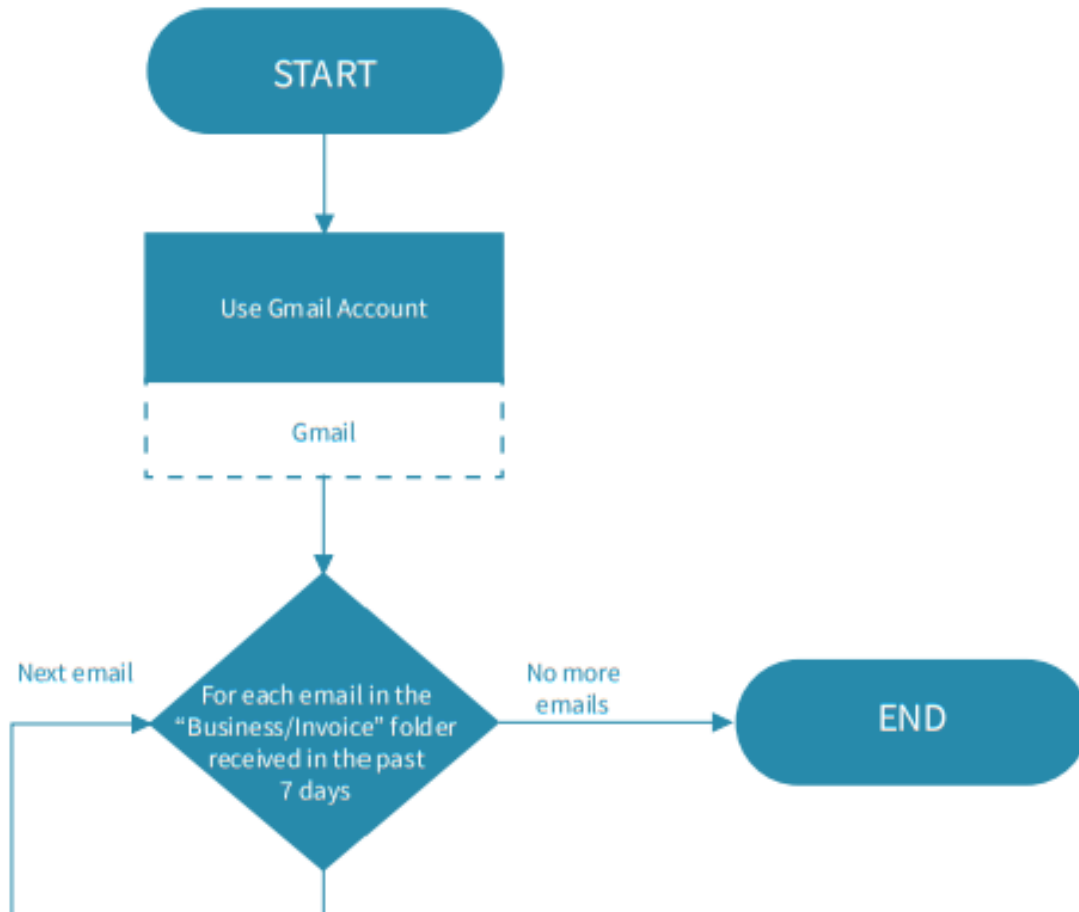
C:\Temp

☐ Overwrite existing

Iterating Through Outlook Mails and Saving Attachments

- Click **Save** in the StudioX ribbon to save the automation, then click **Run** to execute the automation.
- Attachments from each email in the Outlook folder Temp are saved to the Temp folder on the computer.

Iterating Through Outlook Mails and Saving Attachments



Excel Automation with UiPath StudioX

- UiPath StudioX empowers you to automate repetitive tasks within Microsoft Excel, streamlining workflows and enhancing efficiency. Here's a breakdown:
- **What it is:** Excel automation leverages software robots (called **activities** in UiPath) to perform actions within Excel sheets, eliminating the need for manual intervention.
- **How it works:** You visually design workflows using drag-and-drop activities, mimicking human actions like data entry, copying, formatting, calculations, and more.

Excel Automation Benefits

- **Reduces human error:** Automating repetitive tasks minimizes the risk of errors that often occur with manual data manipulation.
- **Saves time and effort:** Freeing employees from tedious Excel tasks allows them to focus on more strategic work.
- **Improves accuracy and consistency:** Automating processes ensures consistent and reliable outcomes.
- **Increases productivity:** By automating time-consuming tasks, overall productivity is enhanced.

Importance and Business Applications:

Excel automation is a valuable tool for businesses across various industries due to:

- **Cost savings:** Reduced manual labor leads to lower operational costs.
- **Faster process execution:** Automating tasks significantly speeds up Excel-based workflows.
- **Enhanced data quality:** Automating data manipulation minimizes errors and ensures data integrity.

Examples of Excel Automation with UiPath StudioX:

- **Data Extraction and Cleaning:**

- Automatically extract data from various sources (websites, databases) into Excel.
- Cleanse and format extracted data, addressing inconsistencies and preparing it for analysis.

- **Report Generation:**

- Populate pre-designed Excel templates with data from different sources.
- Create visually appealing and informative reports with charts, graphs, and formatting.
- Schedule automated report generation at specific intervals.

- **Data Validation and Error Correction:**

- Implement automated checks to identify errors or inconsistencies in large datasets.
- Trigger corrective actions based on defined rules, such as notifying personnel or highlighting errors for manual review.

Examples of Excel Automation with UiPath StudioX:

- **Financial Modeling and Forecasting:**

- Automate calculations and data updates within financial models.
- Create dynamic forecasts based on changing market conditions or business parameters.

- **Inventory Management:**

- Automate stock level tracking and reorder point calculations.
- Generate purchase orders or send alerts when inventory levels reach predefined thresholds.

- StudioX offers a deep integration with Microsoft Excel and ships with a number of activities that automate the tasks you perform in Excel.
- Add an Excel file to automate as a resource to the automation project, and then add the activities to perform in the file.
- When configuring activities that use file, select sheets, tables, named ranges, and named cells directly from **Plus** menu in StudioX.
- Switch from StudioX to Excel to indicate the data to use directly from the file.

About Excel Automation

The screenshot displays the UiPath StudioX interface for a new task named 'NewBlankTask7'. The top ribbon includes tabs for HOME and DESIGN. The DESIGN tab is active, showing a workflow with the following steps: Project, Project Notebook, Manage Packages, App/Web Recorder, Data Extraction, Analyze, Export to Excel, Publish, and Run. The main workspace shows a sequence of activities: Main > SequenceX > Use Excel File > Do > Write Cell. The 'Write Cell' activity is highlighted, and its configuration window is open. The configuration window has a title bar 'Write Cell' and a red exclamation mark icon. It contains two input fields: 'What to write' and 'Where to write', both with a red exclamation mark icon and a plus sign. Below these fields is a checkbox labeled 'Auto increment row'. The left sidebar shows the 'Activities' pane with a search bar and a list of activities categorized by resource (Word, App/Web, Excel, CSV, Outlook, File, Common). The 'Excel' category is selected, showing activities like 'Use Excel File', 'Append Range', 'Auto Fill', 'Copy Paste Range', 'Create Pivot Table', 'Delete Column', 'Delete Rows', 'Delete Sheet', and 'Duplicate Sheet'. The bottom status bar indicates 'Orchestrator Not Connected'.

UiPath StudioX - NewBlankTask7

HOME DESIGN

Save Cut Copy Paste Project Project Notebook Manage Packages App/Web Recorder Data Extraction Analyze Export to Excel Publish Run

Activities Search activities (Ctrl+Alt+F)

Resources

- Use Excel File
Open or create an...

Actions

- Append Range
Copies the data in...
- Auto Fill
Fills cells with data...
- Copy Paste Range
Copies a range or...
- Create Pivot Table
Creates a pivot ta...
- Delete Column
Deletes a specifie...
- Delete Rows
Deletes the specifi...
- Delete Sheet
Deletes a specifie...
- Duplicate Sheet
Creates a copy of...

Main > SequenceX > Use Excel File > Do > Write Cell

Expand All Collapse All

Properties Data Manager

Write Cell

What to write
Select a value

Where to write
Select a value

☐ Auto increment row

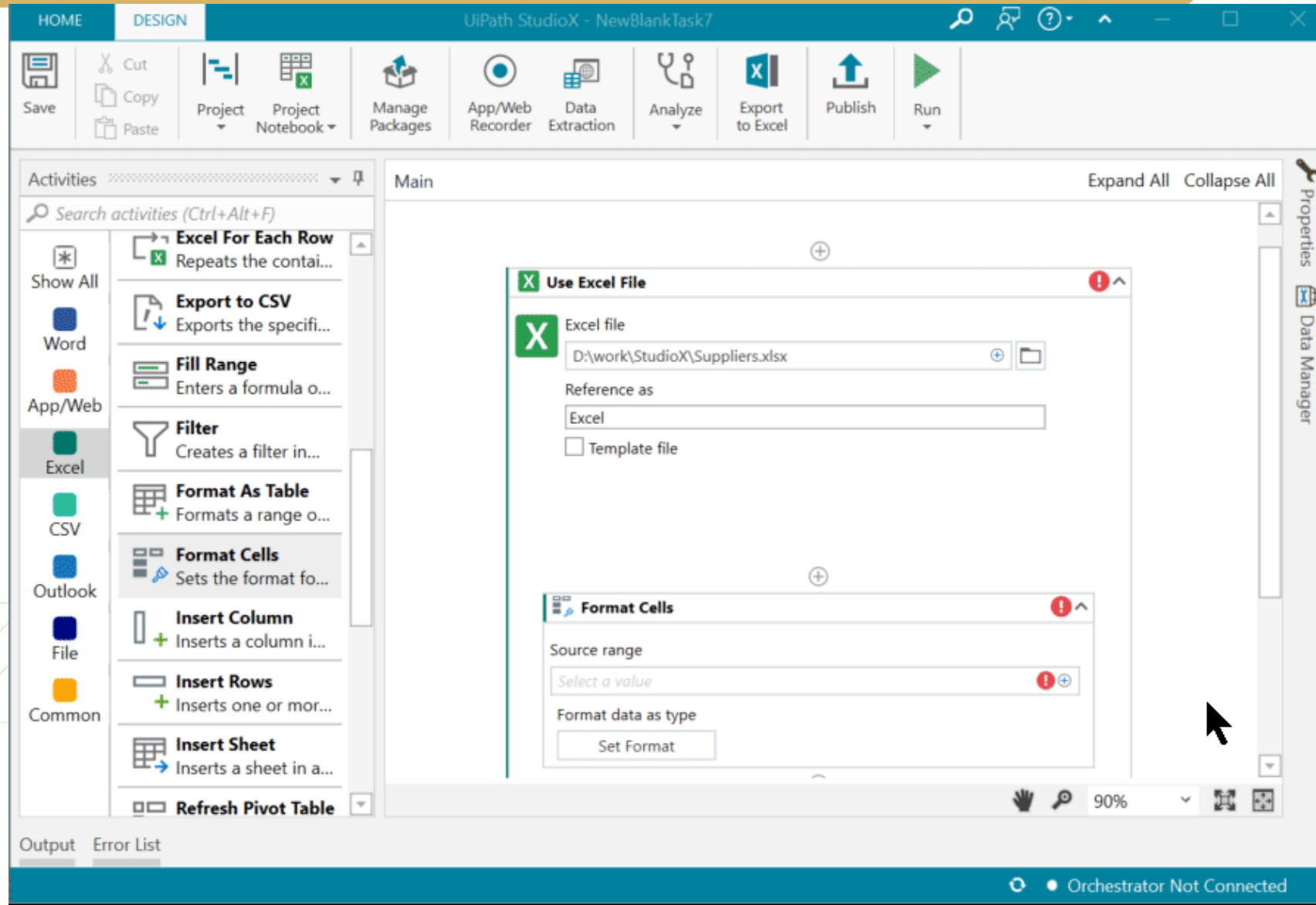
Output Error List

Orchestrator Not Connected



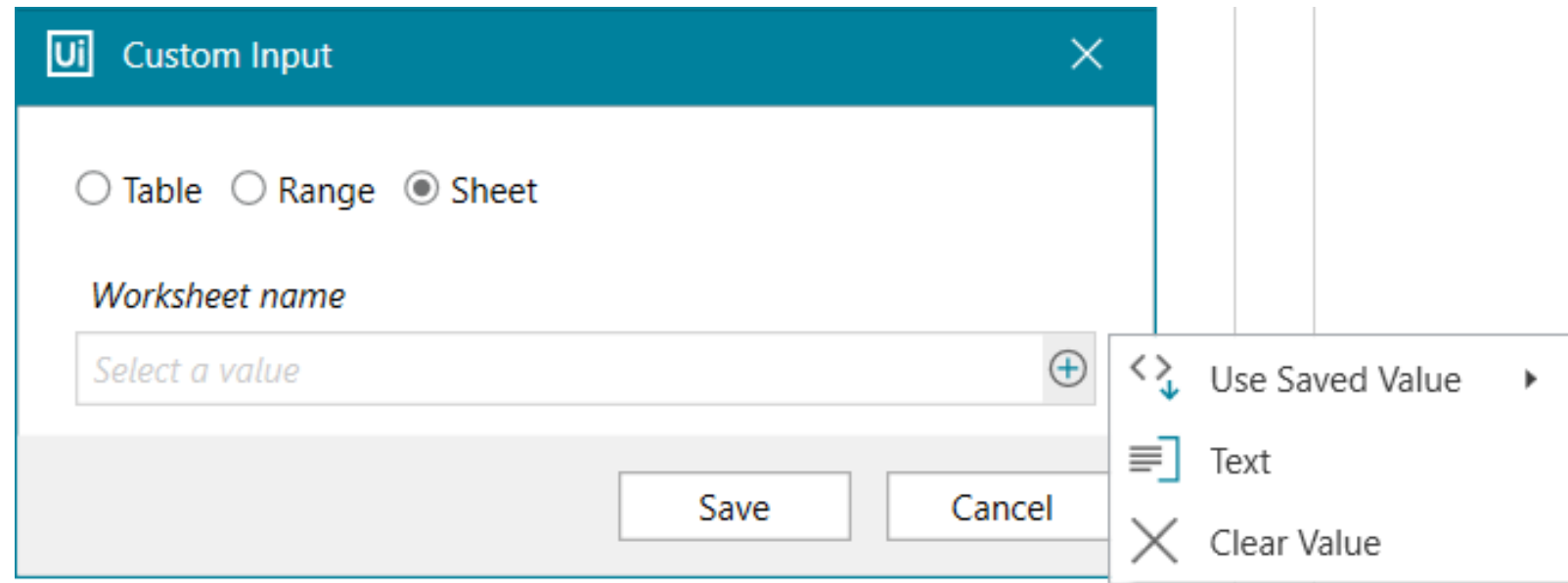
- Click Indicate in Excel in the Plus menu when configuring an activity
- Select data directly from the file, your spreadsheet is opened in Excel and
- UiPath tab appears in the Excel ribbon.
- Select data you want to use, for example a cell or a range of cells, and then click Confirm in the UiPath tab to confirm the selection

About Excel Automation



About Excel Automation

- Manually indicate Excel data to use by selecting Custom Input option in the Plus menu.
- Enter cell and range references, table names, or sheet names, which can be useful when you are working with Excel files with dynamic names, or with files that don't exist.



- There is an Excel file with data about suppliers and find out how many suppliers are there in each city and what number of employees the suppliers have in each city.
- To achieve this, build an automation that creates a pivot table

Step 1: Set up:

- Create a new project.
- Download the provided archive and extract the Suppliers.xlsx file to your project folder.

Step 2: Add and configure Excel file:

- Use the Use Excel File activity to reference Suppliers.xlsx as Suppliers.

Step 3: Create Pivot Table:

- Add the Create Pivot Table activity within Use Excel File.
- Click **Plus** on the right side of the **Source** field, and then select **Suppliers > Indicate in Excel**.
- b) In the Excel file, select all the columns in the **Input** sheet and then click **Confirm** in the UiPath tab in the Excel ribbon.
- You have indicated that you want to create a pivot table for columns A-L in the sheet named Input in the Excel file referenced as Suppliers. The Table range field displays your selection as [Suppliers] Input!A:L.
- c) Click **Plus** on the right side of the **New table name** field and then select **Text**. In the Text Builder window, enter a name for the pivot table, for example CountByCity and click **Save**.
- d) Click **Plus** on the right side of the **Destination** field and then select **Suppliers > Output [Sheet]**.
- Add fields:
 - City (Rows)
 - Count of Internal Name (Values)
 - Sum of Number of Employees (Values)

Solution

Label Pivot Fields (Optional):
Rename Pivot Field1, Pivot Field2, and Pivot Field3 for clarity (e.g., Pivot Field City, etc.).

1 Use Excel File

Excel file

Suppliers.xlsx

Reference as

Suppliers

☒ Save changes ☒ Create if not exists

Read formatting Same as project

☐ Template file

1.1 Create Pivot Table

Source

[Suppliers] Input!A:L

New table name

CountByCity

Destination

[Suppliers] Output

1.1.1 Pivot Field City

Field

City

Is a

Row

1.1.2 Pivot Field Name

Field

Int...

Is a

Value

Function

Count

1.1.3 Pivot Field Employees

Field

Nu...

Is a

Value

Function

Sum

Add Pivot Table Field

- Create an automation to repeat the following workflow for every row in an Excel table that does not already have a value for the **Transaction**
 - Copy the value from the cell in the **Cash In** column.
 - Paste the value to a field in a desktop application.
 - Copy the transaction number from the desktop application.
 - Paste the value to the cell in the **Transaction #** column.
 - Click a button in the desktop application to submit the data.

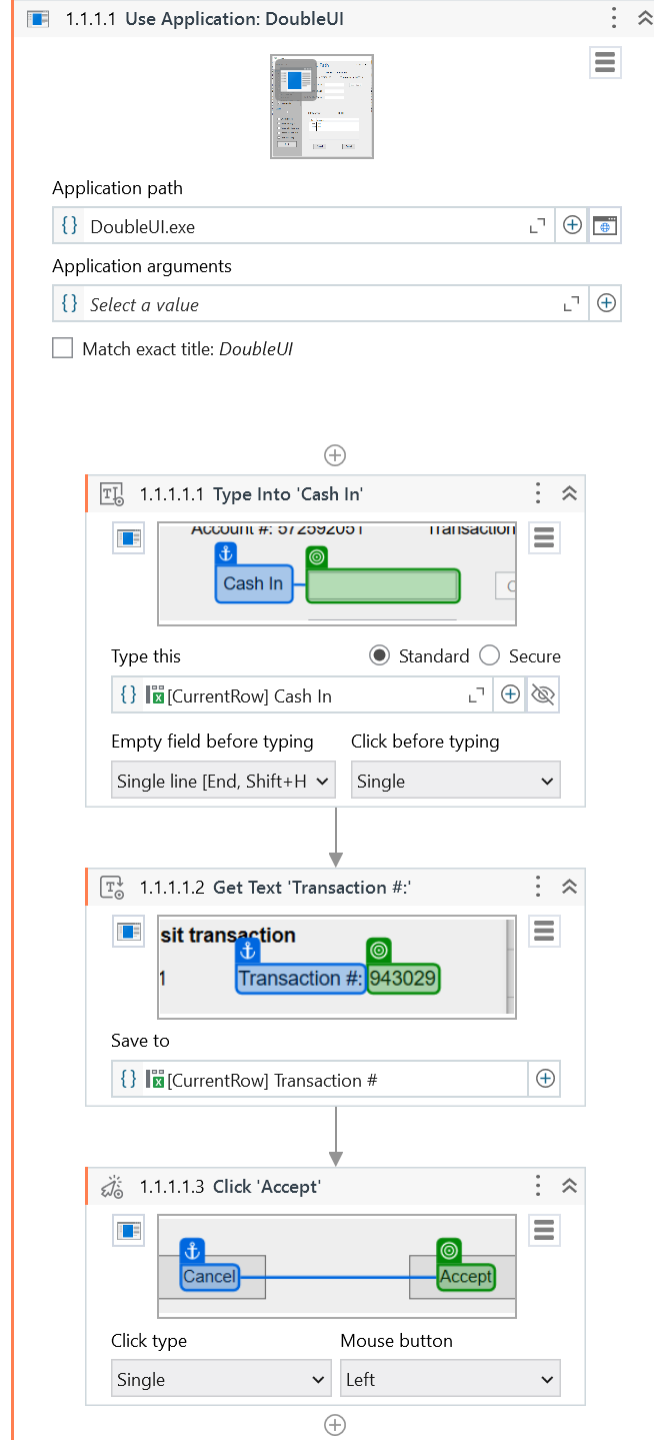
Solution

- **Project Setup:**
 - Create a new project.
 - Download the archive, extract DoubleUI.xlsx, and open DoubleUI.exe.
- **Add Excel File:**
 - Use Use Excel File activity, reference DoubleUI.xlsx as UID.
- **Iterate Through Rows:**
 - Add For Each Excel Row activity and specify the Transactions table in UID.
- **Conditional Execution:**
 - Add an If activity within For Each Excel Row.
 - Check if "Transaction #" in the **current row** is **empty**.
- **Then Branch Actions:**
 - Activities placed here will only run for rows with empty "Transaction #".

The screenshot displays two overlapping windows from a workflow editor. The top window, titled '1 Use Excel File', shows the configuration for an Excel file activity. The 'Excel file' field contains 'DoubleUI.xlsx'. The 'Reference as' field is set to 'UID'. The 'Save changes' and 'Create if not exists' checkboxes are both checked. The 'Read formatting' dropdown is set to 'Same as project'. The 'Template file' checkbox is unchecked. The bottom window, titled '1.1 Excel For Each Row', shows the configuration for an Excel For Each Row activity. The 'For each' field is set to 'CurrentRow'. The 'In range' field is set to '[UID] Transactions'. The 'Has headers' and 'Save after each row' checkboxes are both unchecked. Below this window, a sub-window titled '1.1.1 If' is visible, showing a condition '[CurrentRow] Transaction # is empty' and a 'Then' branch with a placeholder 'Drop Activity Here'.

Solution

- **When a row with an empty "Transaction #" is found:**
- **Open the DoubleUI application:** This step launches the program you'll interact with.
- **Type Cash In value:** The automation finds the "Cash In" field in DoubleUI and enters the value from the current row in the Excel sheet.
- **Get Transaction #:** The automation reads the "Transaction #" value displayed in DoubleUI.
- **Copy Transaction # to Excel:** The automation copies the obtained "Transaction #" and writes it back to the same row in the "Transaction #" column of the Excel sheet.
- **Click "Accept" button:** The automation clicks the "Accept" button within DoubleUI to potentially complete the action.
- **After processing all rows:**
- **Save the automation:** Save your work to preserve the steps you've designed.
- **Run the automation:** Execute the entire process to see it work on your data.



- **Create a project with the following activities:**
 - Two [Use Excel File](#) activities to indicate the Excel file with the invoices and the file where we want to reconcile the data.
 - A series of [Write Cell](#) activities to add two column headers to the reconciliation Excel file and to calculate the sum of invoices for one supplier and the difference between that sum and the amount in the initial summary. We will add a [Copy Range](#) activity to copy the formulas we added for one supplier to all the rows in the sheet so that we can perform the same calculations for all the suppliers, and then add one more Write Cell activity to add the total difference to a cell.
 - An [If](#) activity where we will add a condition that if there are reconciliation errors (the sum of differences is greater than 0), a CSV file with details should be created and emailed to a specified address. For the email tasks, inside the If activity we will add a [Use Desktop Outlook App](#) activity to indicate the Outlook account we want to send the mail from and a [Send Email](#) activity inside Use Desktop Outlook App.

Solution

1. Setting Up:

Work with three Excel files: "invoices.xlsx," "reconcile.xlsx," and "summary.csv."

2. Copying Data:

Data from "invoices.xlsx" and "summary.csv" is copied into separate sheets within "reconcile.xlsx."

3. Calculations:

Formulas are added to the "reconcile.xlsx" file to:

- Compare data between the sheets.

- Calculate differences between corresponding values.

4. Sending Email (if needed):

An email is sent only if the calculated differences don't match (indicating a potential issue).

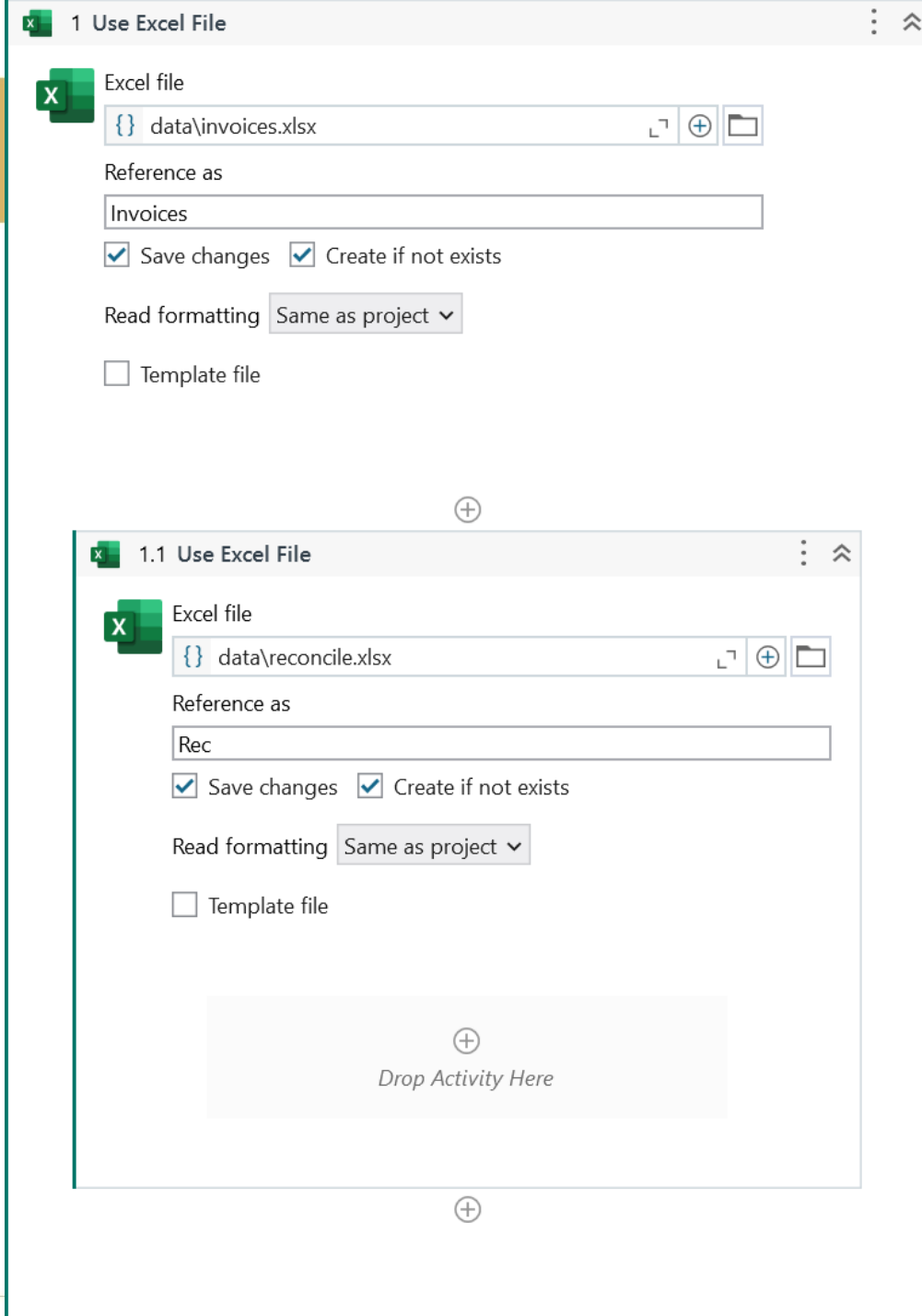
The email includes:

- A CSV file containing details from the "reconcile.xlsx" sheet.

- A subject line mentioning the date.

- A message body highlighting potential errors.

Dr. Divya Agarwal



1.1 Use Excel File

Excel file
data\reconcile.xlsx

Reference as
Rec

☒ Save changes ☒ Create if not exists

Read formatting Same as project

☐ Template file

1.1.1 Read CSV

Read from file
data\summary.csv

Output to
[Rec] Summary

☐ Has headers

1.1.2 Copy Paste Invoices to Rec

Source
[Invoices] Invoices

Destination
[Rec] Invoices

What to copy
All

☐ Exclude source headers ☐ Transpose

1.1.3 Write Cell Header Column 1

What to write
Calculated Amount

Where to write
[Rec] Summary!C1

☐ Auto increment row

1.1.4 Write Cell Sum Column C for Supplier in A2

What to write
=IF(A2 <> "", SUMIF(Invoices!B:B, A2, Invoices!C:C), "")

Where to write
[Rec] Summary!C2

☐ Auto increment row

1.1.5 Write Cell Header Column 2

What to write
Difference

Where to write
[Rec] Summary!D1

☐ Auto increment row

1.1.6 Write Cell C2-B2 if A2 Not Empty

What to write
=IF(A2 <> "", C2-B2, "")

Where to write
[Rec] Summary!D2

☐ Auto increment row

1.1.7 Copy Paste Formula for All Suppliers

Source
[Rec] Summary!C2:D2

Destination
[Rec] Summary!C3:D20

What to copy
Formulas

☐ Exclude source headers ☐ Transpose

1.1.8 Write Cell

What to write
=SUM(Summary!D:D)

Where to write
[Rec] Invoices!F1

☐ Auto increment row

Dr. Divya Agarwal

67

ENGINEERING AND TECHNOLOGY

Solution

1.1.9 If

Condition *

[Rec] Invoices!F1 greater than 0

Then

1.1.9.1 Write CSV

Write from

[Rec] Summary

Write to what file

reconcile- [Notes] Date!YYYYMMDD .csv

☐ Include headers

1.1.9.2 Use Outlook Account

Account

Default Email Account

Reference as

Outlook

1.1.9.2.1 Send Outlook Email

Account

Outlook

To

someone@somewhere.com

Cc

Select a value

Subject

Reconciliation errors for [Notes] Date!Today

Body ☐ HTML ☒ Text ☐ Word Document

One or more suppliers have reconciliation errors. See attache...

☒ Save as draft

Attachments ☒ Files ☐ Folder

reconcile- [Notes] Date!YYYYMMDD .csv

Select a value

Third Unit Complete

- **UNIT III**
- **Error Handling:** Use StudioX tools to validate and analyze automation projects for handling and troubleshooting errors, build automations using best practices that increase reusability and readability.
- **Email Automation:** Use actions and resources related to email automation, create automation projects using StudioX email specific activities, create email content using both text and HTML options.
- **Excel Automation:** Automate tasks involving use of Excel files such as usage of cell activities, range activities, pivot activities, chart activities and workbook activities.